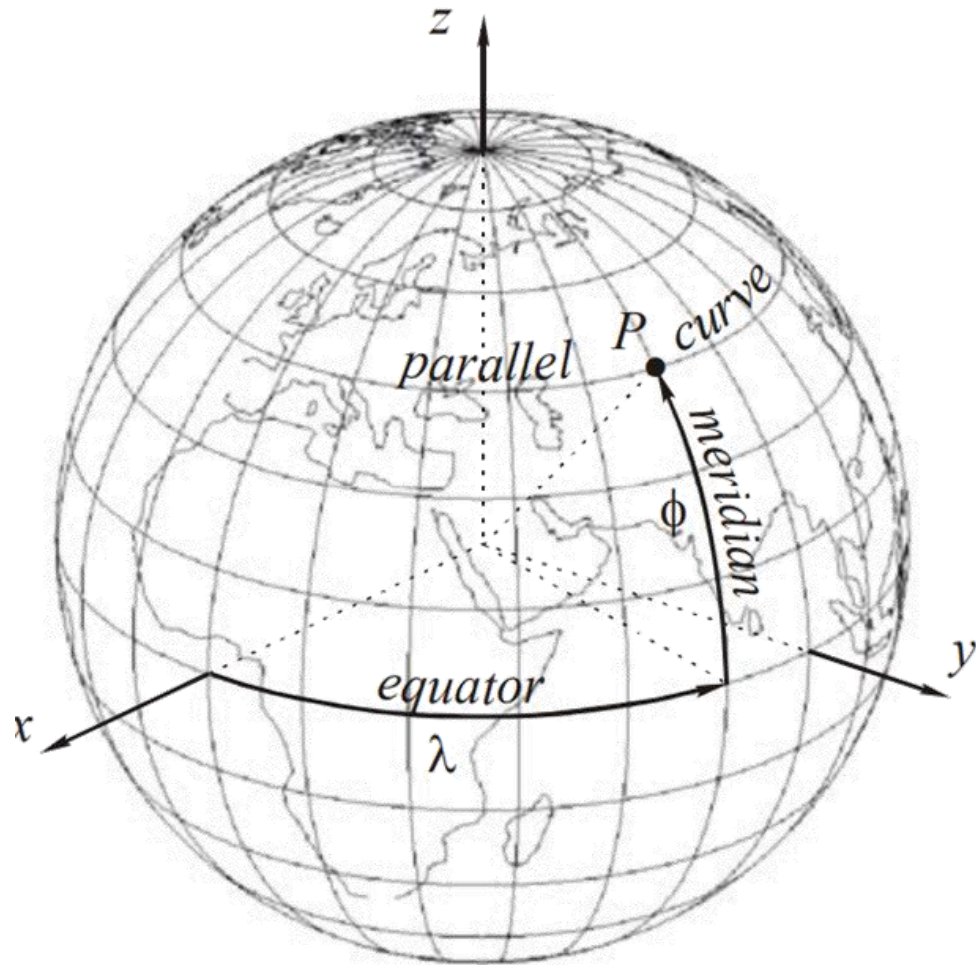


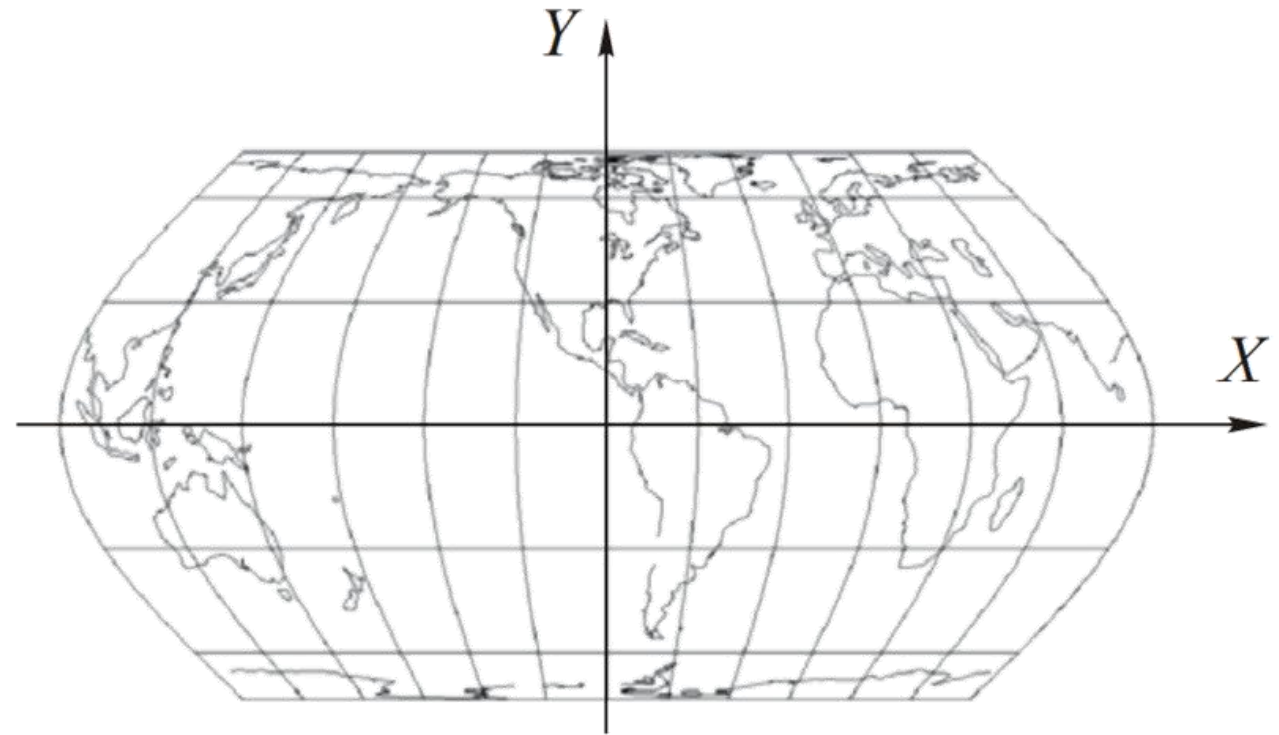
Map Projection Theory

Phisan Santitamnont

Geographic Coord.



Projected Coord.



x, y, z Cartesian vs ϕ, λ curvilinear

$$x = f_1(\phi, \lambda) = R \cos \phi \cos \lambda$$

$$y = f_2(\phi, \lambda) = R \cos \phi \sin \lambda$$

$$z = f_3(\phi, \lambda) = R \sin \phi$$

map projection

$$X = F_1(U, V)$$

$$Y = F_2(U, V)$$

$$Z = F_3(U, V)$$

$$X = \bar{f}_1(\phi, \lambda)$$

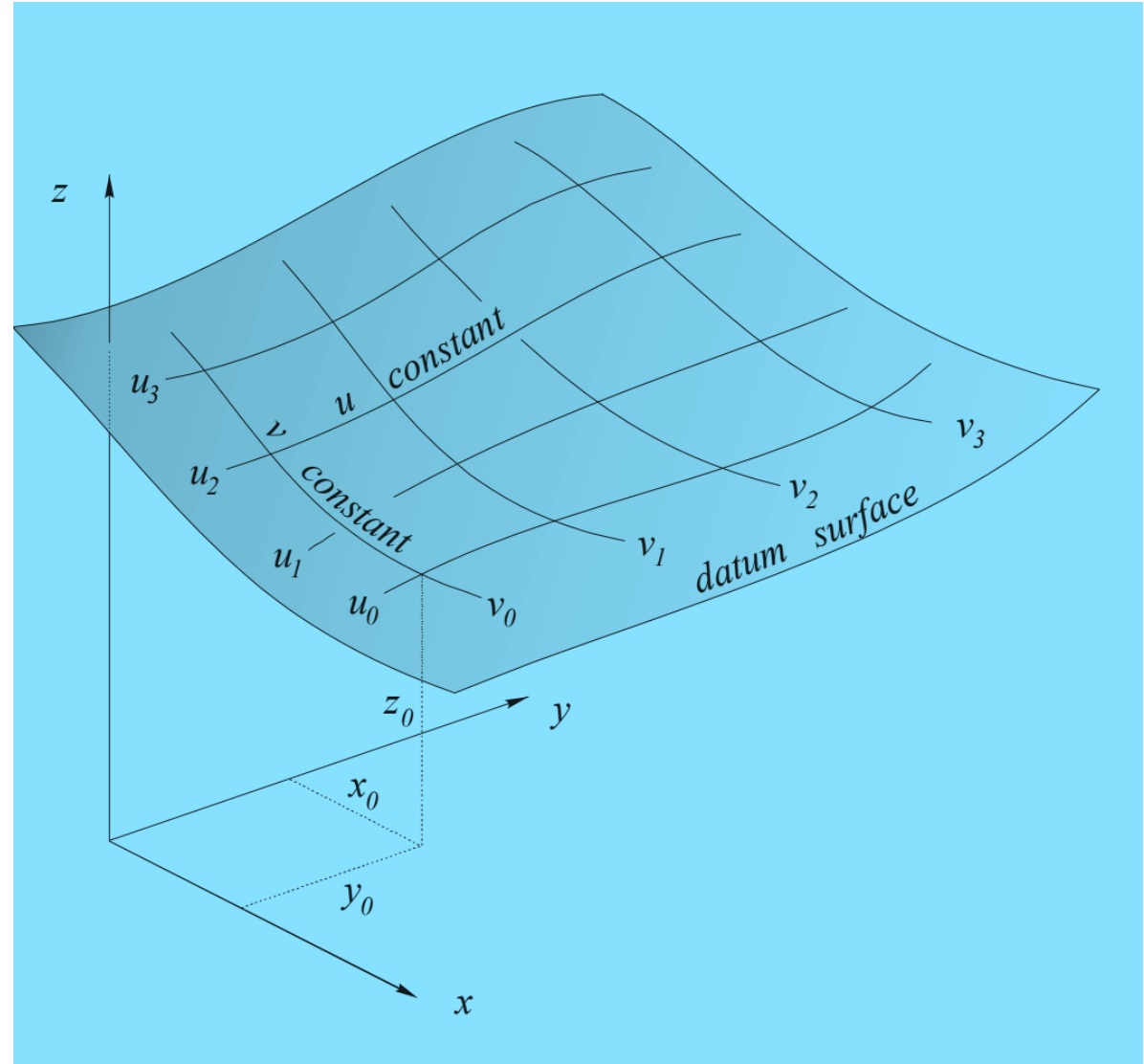
$$Y = \bar{f}_2(\phi, \lambda)$$

Datum Surface

$$x = f_1(u, v)$$

$$y = f_2(u, v)$$

$$z = f_3(u, v)$$

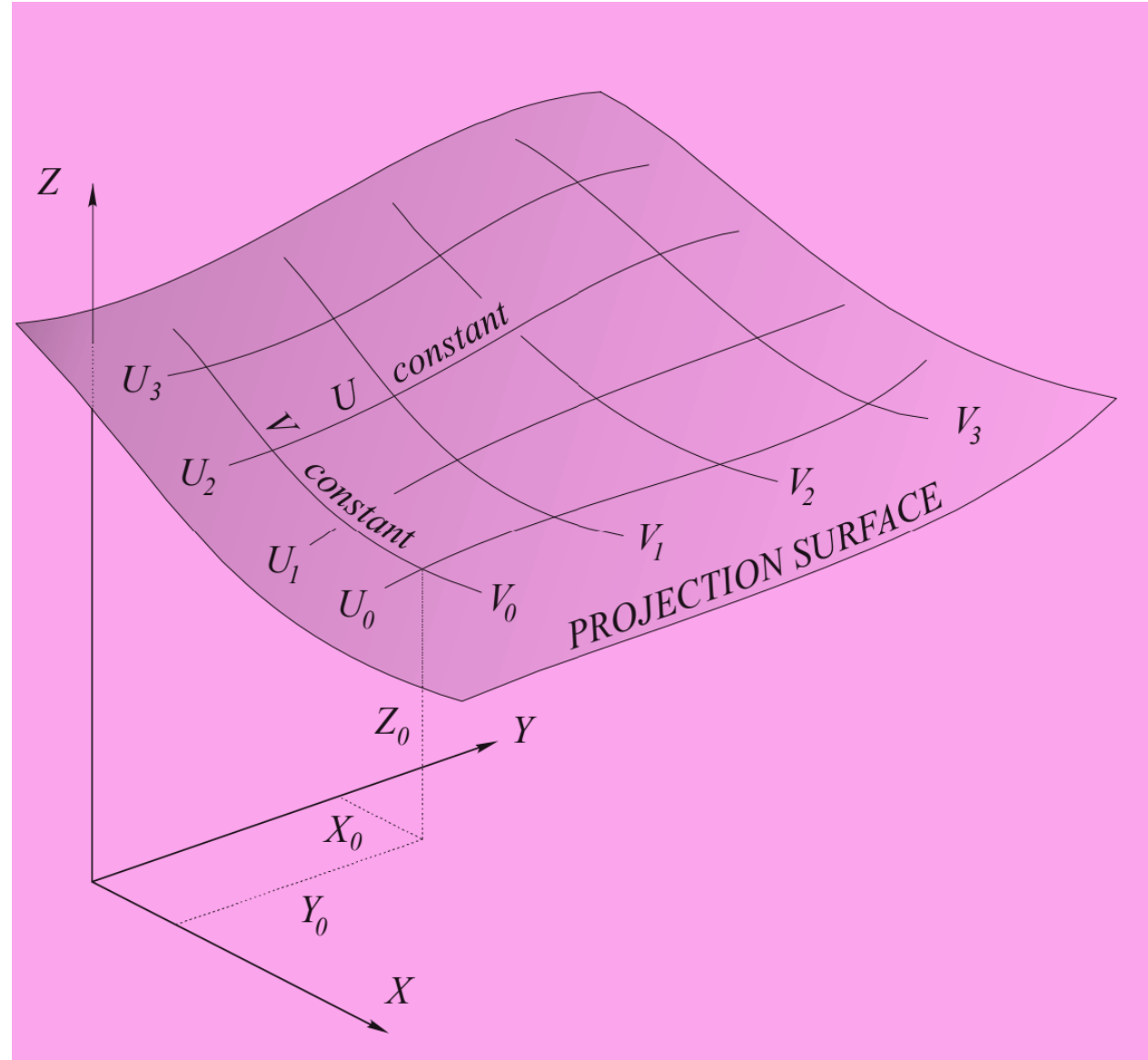


Projection Surface

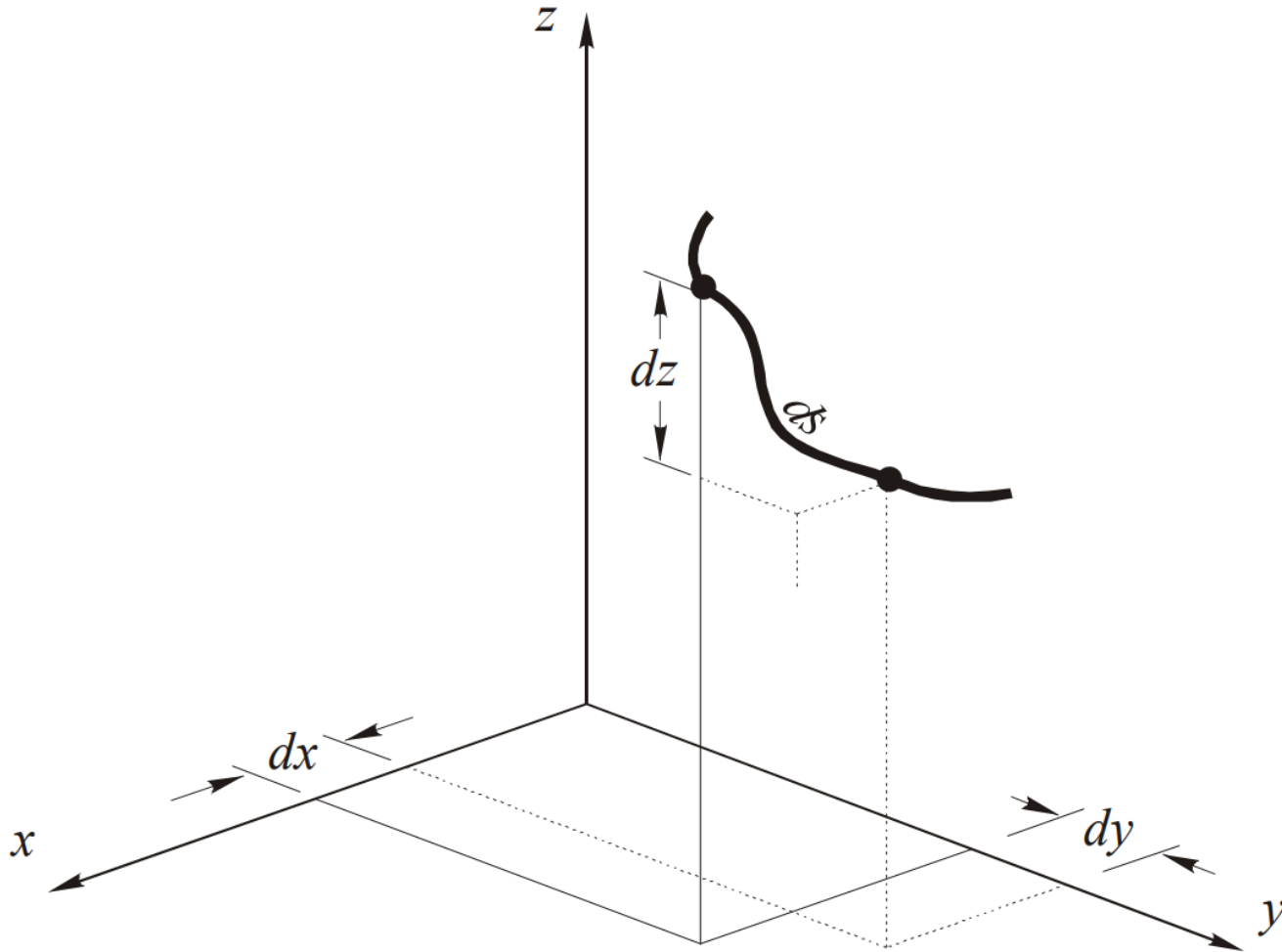
$$X = F_1(U, V)$$

$$Y = F_2(U, V)$$

$$Z = F_3(U, V)$$



THE GAUSSIAN FUNDAMENTAL QUANTITIES ON THE DATUM SURFACE



From differential geometry, the square of the length of a differentially small part of a curve on the datum surface is

$$ds^2 = dx^2 + dy^2 + dz^2 \quad (3.1)$$

$$dx = \frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv$$

$$dy = \frac{\partial y}{\partial u} du + \frac{\partial y}{\partial v} dv$$

$$dz = \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv$$

$$\begin{aligned} ds^2 = & \left(\frac{\partial x}{\partial u} \right)^2 du^2 + \left(\frac{\partial x}{\partial v} \right)^2 dv^2 + 2 \frac{\partial x}{\partial u} \frac{\partial x}{\partial v} du dv \\ & + \left(\frac{\partial y}{\partial u} \right)^2 du^2 + \left(\frac{\partial y}{\partial v} \right)^2 dv^2 + 2 \frac{\partial y}{\partial u} \frac{\partial y}{\partial v} du dv \\ & + \left(\frac{\partial z}{\partial u} \right)^2 du^2 + \left(\frac{\partial z}{\partial v} \right)^2 dv^2 + 2 \frac{\partial z}{\partial u} \frac{\partial z}{\partial v} du dv \end{aligned}$$

$$\begin{aligned} ds^2 = & \left\{ \left(\frac{\partial x}{\partial u} \right)^2 + \left(\frac{\partial y}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial u} \right)^2 \right\} du^2 \\ & + \left\{ \frac{\partial x}{\partial u} \frac{\partial x}{\partial v} + \frac{\partial y}{\partial u} \frac{\partial y}{\partial v} + \frac{\partial z}{\partial u} \frac{\partial z}{\partial v} \right\} 2 du dv \\ & + \left\{ \left(\frac{\partial x}{\partial v} \right)^2 + \left(\frac{\partial y}{\partial v} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 \right\} dv^2 \end{aligned}$$

Gaussian Fundamental Quantities

$$ds^2 = e du^2 + 2f du dv + g dv^2$$

$$e = \left(\frac{\partial x}{\partial u} \right)^2 + \left(\frac{\partial y}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial u} \right)^2$$

$$f = \frac{\partial x}{\partial u} \frac{\partial x}{\partial v} + \frac{\partial y}{\partial u} \frac{\partial y}{\partial v} + \frac{\partial z}{\partial u} \frac{\partial z}{\partial v}$$

$$g = \left(\frac{\partial x}{\partial v} \right)^2 + \left(\frac{\partial y}{\partial v} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2$$

Elemental distance on the surface of the spherical Earth

$$x = f_1(\phi, \lambda) = R \cos \phi \cos \lambda$$

$$y = f_2(\phi, \lambda) = R \cos \phi \sin \lambda$$

$$z = f_3(\phi, \lambda) = R \sin \phi$$

$$e = \left(\frac{\partial x}{\partial \phi} \right)^2 + \left(\frac{\partial y}{\partial \phi} \right)^2 + \left(\frac{\partial z}{\partial \phi} \right)^2$$

$$f = \frac{\partial x}{\partial \phi} \frac{\partial x}{\partial \lambda} + \frac{\partial y}{\partial \phi} \frac{\partial y}{\partial \lambda} + \frac{\partial z}{\partial \phi} \frac{\partial z}{\partial \lambda}$$

$$g = \left(\frac{\partial x}{\partial \lambda} \right)^2 + \left(\frac{\partial y}{\partial \lambda} \right)^2 + \left(\frac{\partial z}{\partial \lambda} \right)^2$$

Gaussian Fundamental Quantities for Sphere

$$\frac{\partial x}{\partial \phi} = -R \sin \phi \cos \lambda$$

$$\frac{\partial y}{\partial \phi} = -R \sin \phi \sin \lambda$$

$$\frac{\partial x}{\partial \phi} = R \cos \phi$$

$$\frac{\partial x}{\partial \lambda} = -R \cos \phi \sin \lambda$$

$$\frac{\partial x}{\partial \lambda} = R \cos \phi \cos \lambda$$

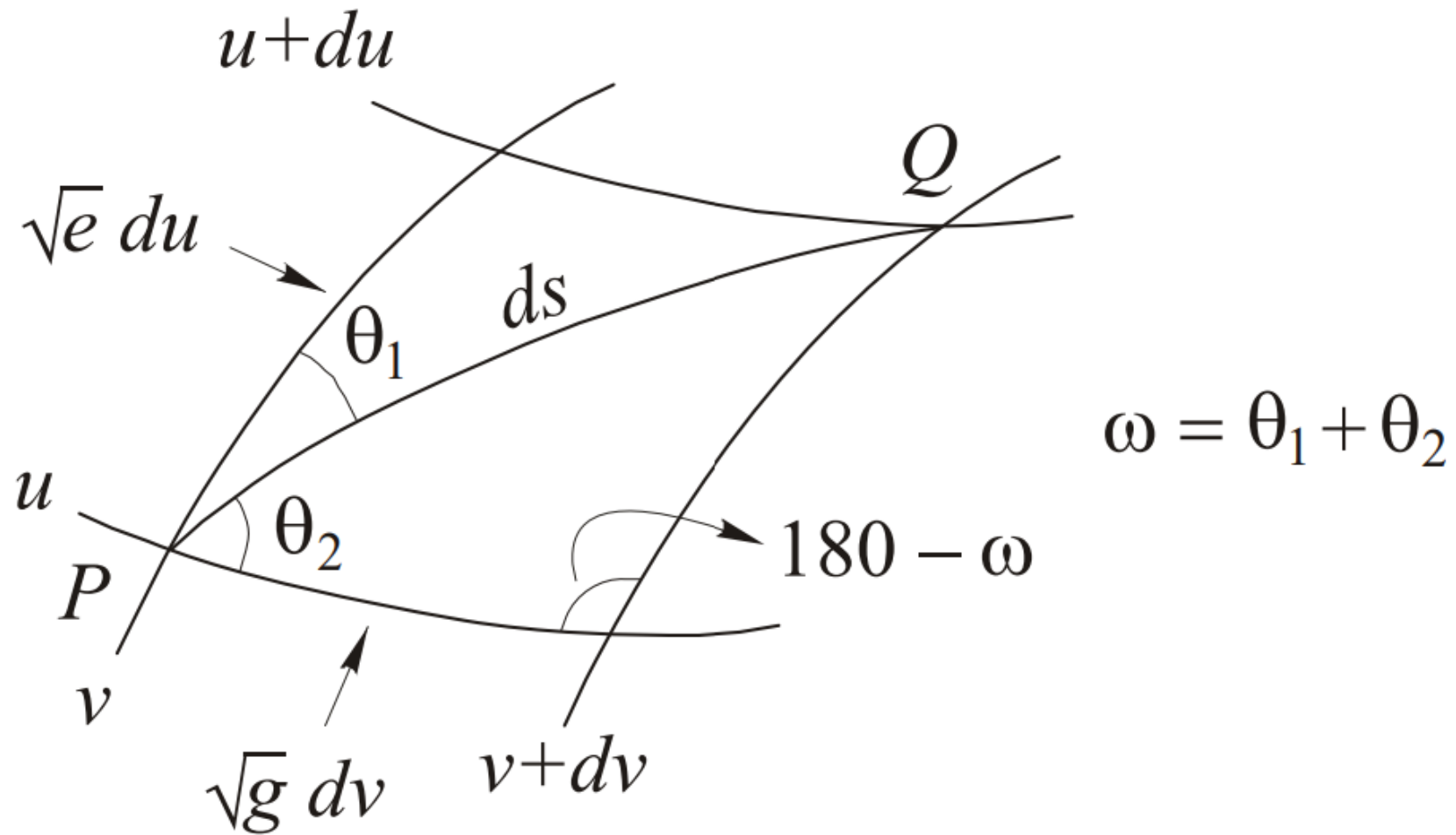
$$\frac{\partial x}{\partial \lambda} = 0$$

$$e = R^2$$

$$f = 0$$

$$g = R^2 \cos^2 \phi$$

PARALLELOGRAM ON THE DATUM SURFACE



distance along parametric curves

$$\begin{aligned} ds_v^2 &= e du^2 + 2f du dv + g dv^2 \\ &= e du^2 + 2f du(0) + g(0)^2 \\ &= e du^2 \end{aligned}$$

$$ds_v = \sqrt{e} du$$

$$\begin{aligned} ds_u^2 &= e du^2 + 2f du dv + g dv^2 \\ &= e(0)^2 + 2f(0)dv + g dv^2 \\ &= g dv^2 \end{aligned}$$

$$ds_u = \sqrt{g} dv$$

angle between parametric curves on the datum surface

$$\begin{aligned} ds^2 &= e du^2 + g dv^2 - 2(\sqrt{e} du)(\sqrt{g} dv)\cos(180 - \omega) \\ &= e du^2 + g dv^2 + 2\sqrt{eg} du dv \cos \omega \end{aligned}$$

$$\cos \omega = \frac{f}{\sqrt{eg}}$$

$$\sin \omega = \frac{j}{\sqrt{eg}}$$

area on the datum surface

$$\begin{aligned} da &= \left(\sqrt{e} du \right) \left(\sqrt{g} dv \right) \sin \omega \\ &= \sqrt{eg} \sin \omega du dv \end{aligned}$$

$$da = j du dv$$

- Note:
- (i) The Gaussian Fundamental Quantities pertaining to the datum surface (x, y, z as functions of u and v) are denoted by e, f and g .
 - (ii) The Gaussian Fundamental Quantities pertaining to the projection surface (X, Y, Z as functions of U and V) are denoted by $\bar{E}, \bar{F}$ and $\bar{G}$.
 - (iii) The Gaussian Fundamental Quantities pertaining to the projection surface (X, Y, Z as functions of u and v) are denoted by E, F and G .

Simple comparison

Symbols	Surface	Parameters used
e, f, g	Datum surface, such as an ellipsoid	u, v , commonly ϕ, λ
$\bar{E}, \bar{F}, \bar{G}$	Projection surface	Its own parameters U, V
E, F, G	Projection surface	Datum parameters u, v

Main idea

$$\bar{E}, \bar{F}, \bar{G}$$

and

$$E, F, G$$

refer to the **same projection surface**, but they use different coordinate systems.

A simple analogy is:

- $\bar{E}, \bar{F}, \bar{G}$: describe the map using map coordinates U, V .
- E, F, G : describe the same map using geographic coordinates ϕ, λ .
- e, f, g : describe the original Earth surface using ϕ, λ .

This allows us to compare

$$ds^2 = e du^2 + 2f du dv + g dv^2$$

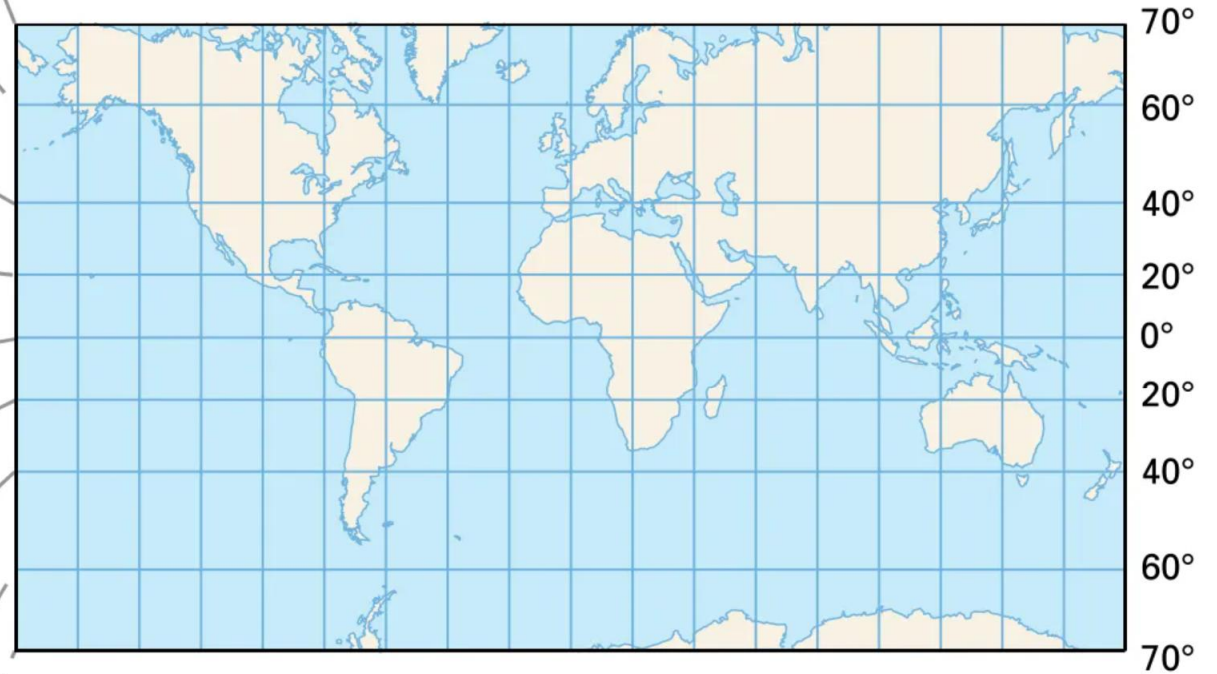
on the Earth with

$$dS^2 = E du^2 + 2F du dv + G dv^2$$

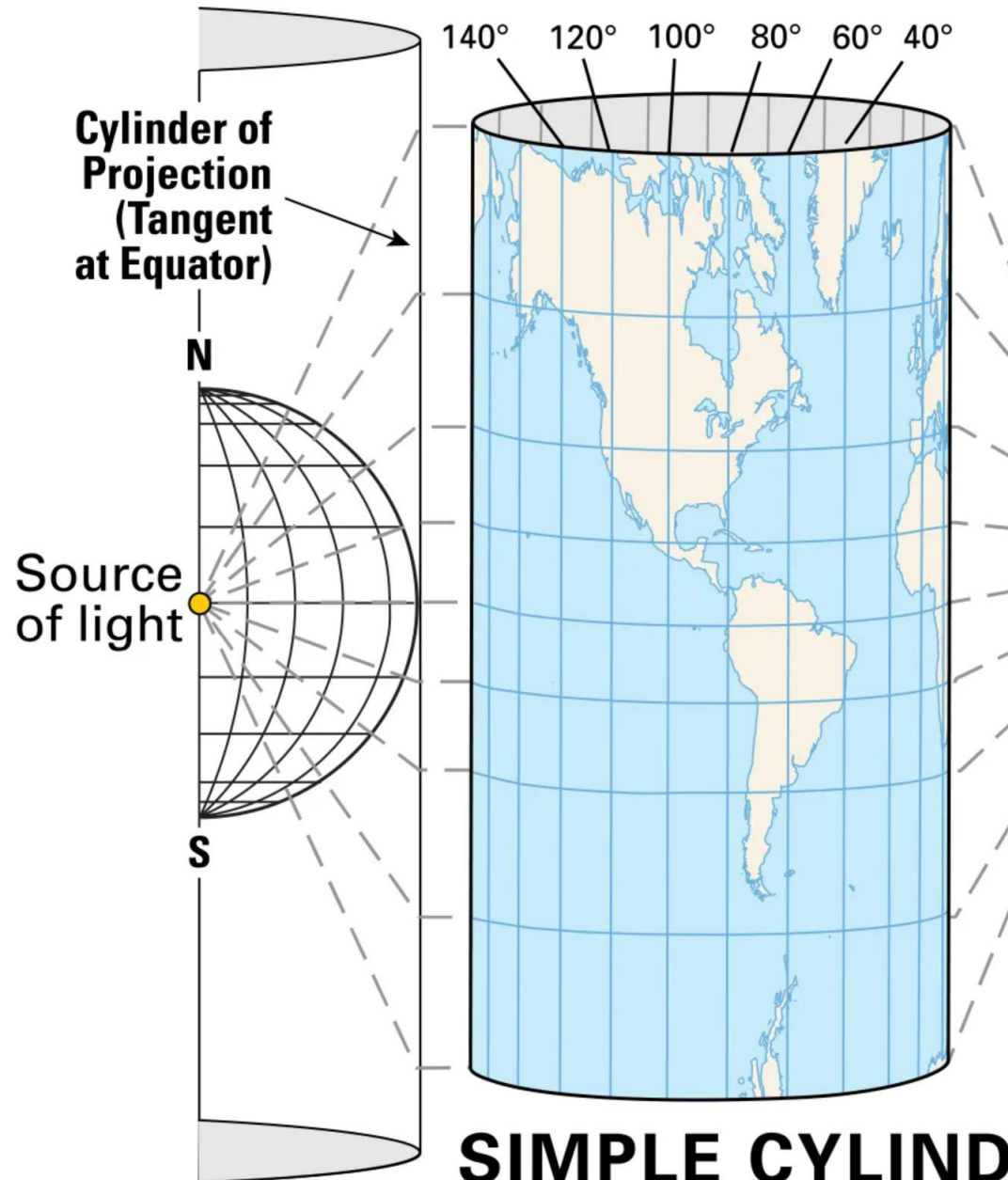
on the map, which is the basis for studying **map scale, angular distortion, and map projection deformation.**

Gaussian Fundamental Quantities and Mercator

MERCATOR PROJECTION

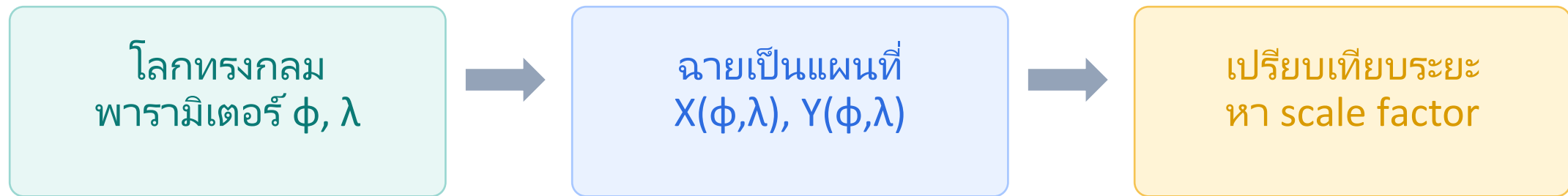


SIMPLE CYLINDRICAL PROJECTION



เปรียบเทียบ e, f, g กับ E, F, G

ตัวอย่างที่ง่ายที่สุด: Simple Mercator บนทรงกลม



แนวคิดหลัก: e, f, g = บนโลก ส่วน E, F, G = บนแผนที่

นิยามสั้น ๆ

ใช้พารามิเตอร์เดียวกัน $u = \phi$, $v = \lambda$ เพื่อเปรียบเทียบโลกกับแผนที่

บนผิวโลก: e, f, g

$$ds^2 = e d\phi^2 + 2f d\phi d\lambda + g d\lambda^2$$

บอกระยะเล็ก ๆ บนทรงกลม/ทรงรีโลก
เหมาะสำหรับอธิบาย “ของจริงบนโลก”

บนแผนที่: E, F, G

$$dS^2 = E d\phi^2 + 2F d\phi d\lambda + G d\lambda^2$$

บอกระยะเล็ก ๆ หลังฉายเป็นแผนที่
ใช้เปรียบเทียบเพื่อหาความผิดเพี้ยน

วิธีจำง่าย : ตัวเล็ก = โลก, ตัวใหญ่ = แผนที่

1) ผิวโลกทรงกลม: e, f, g

สำหรับ Simple Mercator ใช้ตัวอย่างโลกเป็นทรงกลมรัศมี R

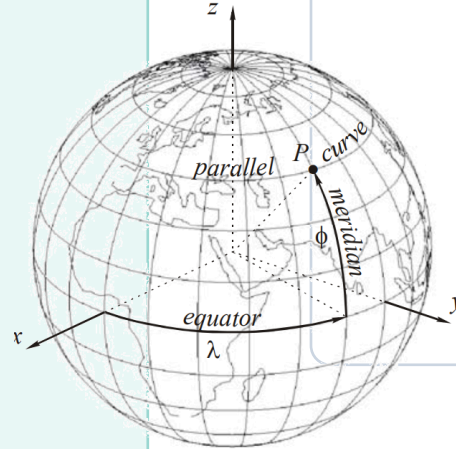
พิกัดบนทรงกลม

$$\begin{aligned}x &= R \cos\phi \cos\lambda \\y &= R \cos\phi \sin\lambda \\z &= R \sin\phi\end{aligned}$$

จากอนุพันธ์ตาม ϕ และ λ

ผลลัพธ์บนผิวโลก

$$\begin{aligned}e &= R^2 \\f &= 0 \\g &= R^2 \cos^2\phi\end{aligned}$$



$$ds^2 = R^2 d\phi^2 + R^2 \cos^2\phi d\lambda^2$$

2) แผนที่ Simple Mercator: E, F, G

สมการฉายแผนที่แบบทรงกลม

สมการฉาย

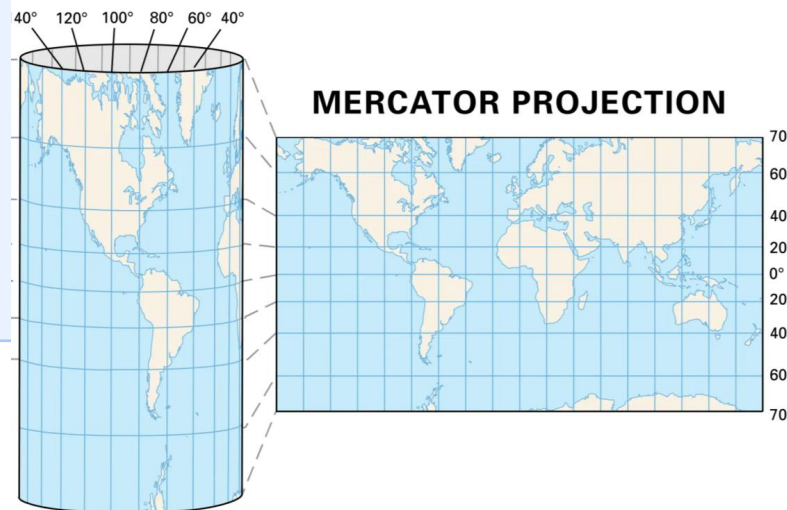
$$\begin{aligned}X &= R\lambda \\Y &= R \ln \tan(\pi/4 + \phi/2) \\Z &= 0\end{aligned}$$

แผนที่เป็นระนาบ จึงใช้ X,Y เท่านั้น

อนุพันธ์ที่ใช้

$$\begin{aligned}X\phi &= 0, & Y\phi &= R \sec\phi \\X\lambda &= R, & Y\lambda &= 0\end{aligned}$$

$$\begin{aligned}E &= R^2 \sec^2\phi \\F &= 0 \\G &= R^2\end{aligned}$$



3) เปรียบเทียบโลกกับแผนที่

นำ e, f, g และ E, F, G มาเทียบกันเพื่อหามาตราส่วน

ปริมาณ	บนโลกทรงกลม	บนแผนที่ Mercator
e หรือ E	$e = R^2$	$E = R^2 \sec^2 \phi$
f หรือ F	$f = 0$	$F = 0$
g หรือ G	$g = R^2 \cos^2 \phi$	$G = R^2$

$$h = \sqrt{E/e} = \sec \phi$$

$$k = \sqrt{G/g} = \sec \phi$$

มาตราส่วนสองทิศเท่ากัน $\rightarrow$ Mercator รักษามุม (conformal)

Simple Mercator เป็นตัวอย่างเริ่มต้นที่ง่ายที่สุด

1. โลก

$$\begin{aligned}e &= R^2 \\f &= 0 \\g &= R^2 \cos^2 \phi\end{aligned}$$

คำนวณง่ายจากทรงกลม

2. แผนที่

$$\begin{aligned}E &= R^2 \sec^2 \phi \\F &= 0 \\G &= R^2\end{aligned}$$

สมการ Mercator สั้น

3. ความหมาย

$$h = k = \sec \phi$$

รักษามุม
แต่ขยายมากเมื่อ ϕ สูง

ประโยคจำ: “ตัวเล็กวัดบนโลก — ตัวใหญ่วัดบนแผนที่”

Ex05 : Gaussian Fundamental Quantities for *Derivation of Gaussian Fundamental Quantities* Spherical **Mercator** Projection

Dr.-Ing. Phisan Santitamnont and AI-LLM

Coordinate System Definitions

Let R be the radius of the spherical Earth.

We define the datum surface parameters u and v as:

- u = geodetic latitude (ϕ), where $u \in [-\pi/2, \pi/2]$
- v = longitude (λ), where $v \in [-\pi, \pi]$

The 3D Cartesian coordinates (x, y, z) on the datum surface (sphere) as functions of u, v are:

$$\begin{aligned}x &= R \cos u \cos v \\y &= R \cos u \sin v \\z &= R \sin u\end{aligned}$$

(i) Datum Surface Quantities: e, f, g

$$\mathbf{r}(u, v) = (x, y, z)$$

Let the position vector be $\mathbf{r}$. We first calculate the partial derivatives and :

$$\mathbf{r}_u \quad \mathbf{r}_v$$

$$\begin{aligned}\mathbf{r}_u &= (-R \sin u \cos v, -R \sin u \sin v, R \cos u) \\ \mathbf{r}_v &= (-R \cos u \sin v, R \cos u \cos v, 0)\end{aligned}$$

Now, calculate the First Fundamental Form coefficients via dot products:

$$\begin{aligned}e &= \mathbf{r}_u \cdot \mathbf{r}_u = (-R \sin u \cos v)^2 + (-R \sin u \sin v)^2 + (R \cos u)^2 \\ e &= R^2 \sin^2 u (\cos^2 v + \sin^2 v) + R^2 \cos^2 u \\ e &= R^2 (\sin^2 u + \cos^2 u) = R^2\end{aligned}$$

(i) Datum Surface Quantities: e, f, g (cont.)

Continuing the calculations for f and g :

$$\begin{aligned} f &= \mathbf{r}_u \cdot \mathbf{r}_v \\ f &= (R^2 \sin u \cos u \sin v \cos v) - (R^2 \sin u \cos u \sin v \cos v) + 0 \\ f &= 0 \end{aligned}$$

$$\begin{aligned} g &= \mathbf{r}_v \cdot \mathbf{r}_v \\ g &= (-R \cos u \sin v)^2 + (R \cos u \cos v)^2 + 0 \\ g &= R^2 \cos^2 u (\sin^2 v + \cos^2 v) \\ g &= R^2 \cos^2 u \end{aligned}$$

Result: $e = R^2, \quad f = 0, \quad g = R^2 \cos^2 u$

(ii) Cartesian Plane Quantities: $\bar{E}, \bar{F}, \bar{G}$

The standard projection surface for the developed Mercator map is a 2D Cartesian plane (unrolled cylinder). Let U be Easting and V be Northing.

$$\begin{aligned} X &= U \\ Y &= V \\ Z &= 0 \end{aligned}$$

Let position vector $\mathbf{R}(U, V) = (U, V, 0)$. The partial derivatives are:

$$\mathbf{R}_U = (1, 0, 0), \quad \mathbf{R}_V = (0, 1, 0)$$

Calculating the fundamental quantities:

$$\bar{E} = \mathbf{R}_U \cdot \mathbf{R}_U = 1^2 + 0 + 0 = 1$$

$$\bar{F} = \mathbf{R}_U \cdot \mathbf{R}_V = 0$$

$$\bar{G} = \mathbf{R}_V \cdot \mathbf{R}_V = 0 + 1^2 + 0 = 1$$

(iii) Projection Surface Quantities: E, F, G

We evaluate the projection plane coordinates (X, Y, Z) using the forward spherical Mercator mapping equations in terms of the datum coordinates u, v :

$$\begin{aligned}X(u, v) &= Rv \\Y(u, v) &= R \ln \tan \left(\frac{\pi}{4} + \frac{u}{2} \right) = R \ln(\sec u + \tan u) \\Z(u, v) &= 0\end{aligned}$$

Let $\mathbf{R}(u, v) = (X(u, v), Y(u, v), 0)$. Calculate partial derivatives for u and v :

For u (latitude):

$$\begin{aligned}X_u &= 0 \\Y_u &= \frac{d}{du} [R \ln(\sec u + \tan u)] = R \left(\frac{\sec u \tan u + \sec^2 u}{\sec u + \tan u} \right) = R \sec u \\ \mathbf{R}_u &= (0, R \sec u, 0)\end{aligned}$$

(iii) Projection Surface Quantities: E, F, G (cont.)

Derivatives for v (longitude):

$$\begin{aligned} X_v &= R, & Y_v &= 0 \\ R_v &= (R, & 0, & 0) \end{aligned}$$

Now calculate E, F, G :

$$\begin{aligned} E &= R_u \cdot R_u = 0^2 + (R \sec u)^2 + 0^2 = R^2 \sec^2 u \\ F &= R_u \cdot R_v = (0)(R) + (R \sec u)(0) + 0 = 0 \\ G &= R_v \cdot R_v = R^2 + 0^2 + 0^2 = R^2 \end{aligned}$$

Result: $E = R^2 \sec^2 u, \quad F = 0, \quad G = R^2$

Gaussian Fundamental for Spherical Mercator

<i>Spherical Datum</i> <i>x,y,z</i> vs atitue <i>u</i> and longitude <i>v</i>	e	f	g
<i>projection surface</i> (Easting, Northing)	$\bar{E}$	$\bar{F}$	$\bar{G}$
<i>Projection surface</i> latitude (ϕ) and longitude (λ),	E	F	G



Ex.	รหัสนิติ ชื่อ-นามสกุล	99	
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Verification of Conformality

Because the Mercator projection is strictly conformal, the scale factor m must be equal in all directions.

We verify this mathematically by checking that the fundamental quantities of the projection surface are strictly proportional to those of the datum surface by a factor of m^2 :

$$\frac{E}{e} = \frac{R^2 \sec^2 u}{R^2} = \sec^2 u$$

$$\frac{G}{g} = \frac{R^2}{R^2 \cos^2 u} = \sec^2 u$$

Since $E/e = G/g = \sec^2 u$ and $F = f = 0$, the derivation confirms the known point scale factor for the spherical Mercator projection:

$$m = \sec u$$

Tissot Indicatrix vs Mercator Projection

